EFY TESTED
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ChatGPT TERMINAL That Talks Using Google TTS

SOMNATH BERA

This project takes the ChatGPT terminal introduced in November 2023 issue to the next level where it now speaks like a VoiceGPT. In this new design, the function of speaking out the questions and answers has been added by incorporating an I2S sound module like the MAX98357A and a 4-ohm loudspeaker. The output power is impressive and distortion-free; it must be heard to be believed.

Two variations are shown here. Fig. 1 displays the author's prototype, and the components required for the terminal are listed in the Bill of Materials (Table 1). Table 2 shows the pin connections between the ESP32 board (MOD1) and the 8.89cm (3.5-inch) TFT (MOD2). Table 3 shows the pin connections between PS2 keyboard and ESP32 board.

Fig. 1: Author's prototype

Circuit and working

Fig. 2 shows the circuit diagram of the ChatGPT terminal that talks to the ESP32 board. It is built around the ESP32 board (MOD1), 8.89cm (3.5-inch) TFT (MOD2), MAX98357A (MOD3), 5V voltage regulator 7805 (IC1), PS2 keyboard, and a few other components.

Connect the components and display according to the circuit diagram. The voltage regulator used in the circuit powers the device in the voltage range of 5V to 9V. However, you can replace the voltage regulator with a 5V regulated power adaptor.

Note that the 'gain' of MAX98357A is connected to ground to increase output power. Removing this connection will slightly reduce the output power.

A PS2 keyboard like a USB key-

TABLE 1
BILL OF MATERIALS

Items	Quantity
ESP32 – MCU (MOD1)	1
PS2 keyboard	1
8.89cm (3.5-inch) TFT (MOD2)	1
Wires, PCB, connector	20
IC 7805: 5V regulator IC (IC1)	1
I2S sound module MAX98357A (MOD3)	1
100MFD capacitor (C1, C2)	2
Rectifier diode 1N4007 (D1)	1

TABLE 2
PIN CONNECTIONS BETWEEN ESP32 BOARD AND 8.89CM (3.5-INCH) TFT

8.89cm (3.5-inch) TFT pin	ESP32 Board Pin	8.89cm (3.5-inch) TFT Pin	ESP32 Pin
LCD_CS	G33	D02	G26
LCD_DC	G15	D03	G25
LED RST	G32	D04	G16/RX2
WR	G4	D05	G17/TX2
RD	G2	D06	G27
D0	G12	D07	G14
D01	G13	VCC/5 Volt	5V regulator
Gnd	Gnd	ESP32 Gnd to be connected to 5V regulator ground and ESP32 Vin to be connected to 5V regulator output	